Energy-Efficient Dual Hydroponic System for Organic/Inorganic Capsicum Chili Farming

25-26J-035

Project Proposal Report

P.A.S.S. WEERASEKARA

BSc (Hons) in Information Technology

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P.A.S.S. Weerasekara – IT22258762

Supervisor: Ms. Shashika Lokuliyana

Co-Supervisor: Mr. Dinith Primal

BSc (Hons) in Information Technology

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Department of Computer System Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
P.A.S.S. Weerasekara	IT22258762	Sapuli

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: ……………… Date: 19/9/25

ABSTRACT

Traditional agricultural practices in Sri Lanka are highly susceptible to climate variability and resource scarcity, while conventional hydroponic solutions, though effective for leafy greens, lack the adaptability required for fruit-bearing crops like capsicum. These systems face significant constraints in automated nutrient control, real-time tracking, and power optimization when deployed in resource-limited, off-grid settings. This study addresses these limitations by providing the design and implementation of a semi-automated, IoT-enabled inorganic Nutrient Film Technique (NFT) hydroponic system tailored for capsicum chili cultivation under Sri Lankan agro-climatic conditions.

The proposed solution leverages a comprehensive multi-sensor network (pH, EC, temperature, humidity) for continuous monitoring, with a cloud-connected dashboard supporting real-time visualization and analytics. A core innovation is the real-time fuzzy logic-based control system for intelligent nutrient and pH management, ensuring dynamic adjustments to environmental fluctuations. Furthermore, the system incorporates a water reuse and recycling mechanism with in-line filtration and sterilization, as well as a semi-automated sensor calibration feature to ensure reliable long-term performance.

This research aims to deliver a scalable, low-maintenance solution that enhances crop yield and minimizes resource wastage by integrating IoT and smart management algorithms. The anticipated outcome is a replicable and scalable hydroponic model that empowers smallholder farmers with accessible precision farming technology, contributing to greater food security and resilience against climatic and economic pressures in Sri Lanka and other developing nations. By bridging current gaps in automation, resource efficiency, and real-world adaptability, the system provides a resilient and sustainable solution for smallholder farmers.

Keywords: NFT hydroponics, fuzzy logic control, IoT monitoring, water recycling, capsicum chili cultivation

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LIST OF ABBREVIATION

Abbreviation	**Full Name**
AIMS	Autonomous Intelligent Machines and Systems
API	Application Programming Interface
EC	Electrical Conductivity
ESP32	Expressif Systems 32
FL	Fuzzy Logic
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
LKR	Sri Lankan Rupee
MQTT	Message Queuing Telemetry Transport
NPK	Nitrogen–Phosphorus–Potassium
NFT	Nutrient Film Technique
pH	Potential of Hydrogen
QoS	Quality of Service
RH	Relative Humidity
TCP/IP	Transmission Control Protocol/Internet Protocol
ToF	Time-of-Flight
Wi-Fi	Wireless Fidelity

1. INTRODUCTION

The rapid increase in global population requires a significant boost in food production, with projections suggesting that food demand will rise by 60% by 2050 [6]. Conventional farming practices are increasingly strained due to challenges such as fluctuating weather patterns, declining soil fertility, and limited water availability. These challenges necessitate the development of sustainable agricultural methods to ensure global food security and economic sustainability [7], [8].

Hydroponics, a groundbreaking method of growing plants without soil, has emerged as a promising alternative, utilizing nutrient-rich water solutions to promote plant growth [9]. This method offers several advantages over traditional farming, including a 90% reduction in water usage due to recycling and recirculation [10]. Additionally, hydroponic systems provide better control over growing conditions, leading to higher crop yields per unit area and more efficient resource utilization. These systems allow for year-round crop cultivation, independent of external weather conditions, and reduce the need for harmful pesticides and herbicides by creating a controlled environment [6], [8], [12].

The Nutrient Film Technique (NFT) is one of the most efficient hydroponic methods, where a thin film of nutrient solution flows over the plant roots, providing them with nutrients, water, and oxygen [13]. NFT systems have proven to enhance growth conditions for a variety of crops, especially those requiring high levels of nutrient control, such as leafy greens and herbs. However, applying NFT for more complex crops such as capsicum chili introduces challenges due to the need for precise nutrient and pH management. Capsicum chili cultivation is highly sensitive to fluctuations in these variables, which can significantly affect growth and yield quality [15].

The integration of Internet of Things (IoT) technology into hydroponic systems has significantly transformed modern farming, resulting in the emergence of "smart farming" solutions. IoT-enabled systems allow real-time monitoring and control of critical environmental factors such as temperature, humidity, pH levels, and nutrient concentrations, which minimizes the need for human intervention [7], [9].

This automation ensures that plants always have access to the required resources, leading to more efficient, accurate, and sustainable farming practices.

This research aims to design and implement an energy-efficient dual hydroponic system for both organic and inorganic capsicum chili farming, with a focus on the inorganic NFT hydroponic system. The proposed system will integrate several innovations, including real-time fuzzy logic-based adaptive nutrient and pH control, sensor-based growth monitoring using ultrasonic and ToF sensors, and water and nutrient recycling with in-line filtration. The use of fuzzy logic will allow the system to dynamically adjust nutrient dosing, pH balancing, and water flow, ensuring optimal conditions for plant growth and reducing human intervention [4], [15].

Furthermore, a cloud-based dashboard will be developed to provide remote monitoring and management capabilities. This dashboard will allow farmers to track system performance, receive alerts, and adjust as necessary, thereby improving efficiency and sustainability in hydroponic farming. The system will also incorporate semi-automated sensor calibration to ensure long-term accuracy and reliability of environmental measurements [14].

By integrating IoT, fuzzy logic, and automated controls, the proposed system will contribute to the sustainability and performance of hydroponic farming systems, specifically for capsicum chili cultivation. It will also address the growing demand for high-quality capsicum chili in Sri Lanka, which has been severely impacted by climate change and soil degradation [19], [20].

1.1 Background

1.1.1 Hydroponics as a Sustainable Alternative in Agriculture

Hydroponic farming presents a viable solution to address the growing challenges faced by traditional agriculture, especially in resource-constrained regions like Sri Lanka. With the increasing impacts of climate change, land degradation, and water scarcity, conventional agricultural methods are becoming unsustainable, particularly in urban areas where space is limited [10], [15]. Hydroponics, as a method of soil-less cultivation using nutrient-rich water, provides a more efficient and sustainable alternative by reducing water consumption, eliminating soil-borne diseases, and enabling year-round crop production [8].

The Nutrient Film Technique (NFT) is one of the most widely used hydroponic methods due to its efficient nutrient delivery system. In this setup, a thin film of nutrient solution continuously flows over plant roots, allowing plants to absorb the necessary nutrients while minimizing water waste [10], [12]. Research has shown that the NFT method not only optimizes space use but also provides significant water savings compared to traditional farming. This is crucial in areas where water resources are limited [13], [14].

Capsicum chili, a widely cultivated crop in Sri Lanka, can thrive in hydroponic systems where environmental factors like pH, temperature, and humidity can be meticulously controlled [16], [13]. In these systems, IoT-enabled sensors monitor key parameters, allowing for automated adjustments that ensure optimal growth conditions [14]. This precise control over the growing environment contributes to higher yields and better-quality produce, addressing the increasing demand for fresh food in urban areas [15], [17].

However, the challenges associated with maintaining nutrient balance, water efficiency, and real-time environmental monitoring persist. While hydroponic farming systems, particularly NFT, have proven successful in overcoming land and water scarcity, they still face issues such as nutrient management and system automation [12], [13]. As a result, integrating advanced technologies can further enhance the sustainability and efficiency of hydroponic systems [10], [18].

1.1.2 Advances in IoT and Automation in Hydroponics

The integration of IoT technologies into hydroponic farming systems has transformed the way environmental conditions are monitored and managed. IoT systems use sensors to collect real-time data on critical parameters such as pH, temperature, humidity, and electrical conductivity (EC), which are essential for optimizing plant growth [15], [16]. This data is transmitted to cloud-based platforms, allowing for remote monitoring and control of hydroponic systems, which is especially beneficial for large-scale or urban farming operations [14], [13].

Several studies have demonstrated the advantages of IoT in hydroponics, particularly in the real-time tracking of environmental variables. A study on lettuce cultivation using IoT-enabled systems showed significant improvements in crop growth and yield due to the continuous monitoring and adjustment of environmental factors such as pH and nutrient concentrations [10], [15]. However, many existing IoT-based hydroponic systems still rely on threshold-based control methods, which fail to dynamically adjust to changing environmental conditions or plant-specific requirements. This limitation can result in suboptimal growth and inefficient resource use [13], [10].

To address these challenges, fuzzy logic controllers have been proposed for integrating real-time adaptive control into hydroponic systems. Fuzzy logic allows systems to make real-time adjustments to nutrient levels and pH based on the continuous input data from IoT sensors [18], [17]. This enables systems to adapt dynamically to variations in plant needs and environmental conditions, ensuring better resource utilization and plant health [10], [12]. Moreover, IoT systems can help automate water management, nutrient dosing, and environmental regulation, reducing the need for manual monitoring and minimizing the risk of human error [14], [17].

Despite the advances in automation and control, there is still room for improvement in achieving real-time dynamic nutrient management and water recycling in hydroponic systems. Future research should focus on refining these systems to ensure they can respond more effectively to rapid changes in environmental conditions, plant health, and nutrient needs [12], [19].

1.2 Literature Review

Hydroponic farming, specifically the Nutrient Film Technique (NFT), has emerged as a sustainable and efficient alternative to traditional soil-based farming. It allows for precise control over environmental factors, ensuring optimal plant growth in resource-constrained environments. With the increasing global demand for food and urbanization, hydroponics, particularly when enhanced with IoT, presents a promising solution to address land degradation, water scarcity, and climate change [6].

1.2.1 IoT Integration in Hydroponic Systems

Recent research has demonstrated significant advancements in IoT-based hydroponic monitoring and control systems. Shaha and Budhathoki developed an IoT-driven automation system for hydroponic lettuce farming, utilizing ESP32 microcontrollers with integrated sensors for pH, electrical conductivity (EC), temperature, and humidity monitoring [1]. Their system achieved real-time data visualization through custom IoT dashboards, demonstrating the feasibility of remote monitoring capabilities. However, their approach relied on conventional threshold-based control mechanisms, limiting the system's ability to adapt dynamically to changing environmental conditions.

Similarly, Bhargava et al. implemented a sensor fusion-based intelligent hydroponic system that integrated multiple IoT devices for comprehensive environmental monitoring [2]. Their work demonstrated power savings of 20.4% during water and temperature regulation and 82.1% during light regulation through the implementation of Random Forest algorithms for priority-based parameter selection. Despite these achievements, the system lacked real-time adaptive control mechanisms and failed to address water recycling or sensor calibration challenges [2], [20].

Wildan and Anisa conducted comprehensive experiments comparing IoT-enabled hydroponic systems with conventional manual methods over a 30-day cultivation period [8]. Their results revealed superior plant health and growth outcomes with IoT implementation, maintaining optimal pH (6.9-7.0) and nutrient levels (1250-1265 ppm) consistently, while conventional methods resulted in plant mortality due to declining environmental conditions. However, their system design did not incorporate advanced control algorithms or predictive analytics capabilities [15].

1.2.2 NFT System Design and Implementation

The Nutrient Film Technique has been extensively researched for its water efficiency and nutrient delivery capabilities. Ali et al. developed a comprehensive NFT hydroponic system specifically optimized for Dwarf Pak Choi cultivation, achieving precise monitoring of electrical conductivity (1.62-1.82 mS/cm), pH (5.938-6.109), and dissolved oxygen (8.465-8.513 mg/L) [12]. Their system demonstrated excellent sensor calibration accuracy with minimal deviations from theoretical values, yet the control mechanisms remained static without adaptive fuzzy logic implementation [7], [12].

Sayekti et al. designed a smart farming system using NodeMCU ESP32 for NFT hydroponics, integrating TDS, pH, and ultrasonic sensors with Blynk cloud platform for remote monitoring [13]. While their system successfully demonstrated automated nutrient and pH control through relay-based actuators, it lacked sophisticated control algorithms and did not address water reuse or filtration mechanisms [18].

Zaid et al. implemented an IoT-based smart hydroponic system using NFT for lettuce cultivation, incorporating multiple sensors and achieving power efficiency improvements through automated scheduling [5]. Their system maintained optimal growing conditions with 87% sensor accuracy for TDS measurements and 99.02% for pH sensors. However, the research did not explore advanced control methodologies or predictive maintenance capabilities [2].

1.2.3 Advanced Control Systems and Machine Learning Integration

Several researchers have explored intelligent control mechanisms for hydroponic systems. Puno et al. developed a fuzzy logic control system for NFT hydroponics, implementing Mamdani fuzzy inference for nutrient and pH regulation [17]. Their approach demonstrated improved response times and reduced steady-state errors compared to conventional PID controllers. However, the system lacked real-time adaptation capabilities and did not integrate cloud-based monitoring or predictive analytics.

Fitriani et al. implemented a Convolutional Neural Network (CNN) approach for nutrition control in NFT hydroponics, achieving error rates of 3.35% for nutrient control and 0.98% for pH control [18]. Their CNN-based system operated entirely within the ESP32 microcontroller, demonstrating autonomous operation without external computing requirements. Despite these achievements, the research did not address sensor calibration automation or water recycling mechanisms.

Suresh and Selvabhuvaneswari developed a machine learning-enhanced hydroponic system using Random Forest algorithms for nutrient optimization, achieving 50% improvements in crop yield and quality [7]. Their comprehensive web-based application integrated real-time monitoring with market analysis capabilities. However, the system lacked fuzzy logic implementation and did not incorporate water reuse mechanisms or advanced sensor calibration protocols [5].

1.2.4 Sensor Technologies and Calibration Challenges

Sensor accuracy and calibration remain critical challenges in hydroponic systems. Zheng et al. conducted extensive research on ultrasonic sensor calibration for crop canopy height measurements, identifying significant factors affecting measurement accuracy including observation angle and planting density [16]. Their calibration model achieved RMSE improvements from 967.3 mm to 87.3 mm for maize and 216.7 mm to 42.0 mm for wheat canopy height measurements. This research highlighted the critical importance of systematic sensor calibration protocols, which most existing hydroponic systems fail to address adequately [16].

Gokulraj et al. developed an IoT-based nutrition alert system focusing on automated nutrient monitoring and control [14]. While their system demonstrated effective real-time monitoring capabilities, it relied primarily on threshold-based alerts rather than predictive analytics or adaptive control mechanisms. The research did not address long-term sensor drift or automated calibration procedures [8].

1.2.5 Cloud Integration and Data Analytics

Cloud-based monitoring and control have become essential components of modern hydroponic systems. Perera et al. developed intelligent algorithms for optimizing hydroponic solutions in IoT-integrated systems, demonstrating improved resource utilization through cloud-based analytics [11]. However, their system lacked comprehensive predictive capabilities and did not integrate advanced control algorithms for real-time adaptation [1].

Lal implemented an autonomous hydroponic farming system using IoT with emphasis on remote monitoring and control capabilities [9]. While the system demonstrated effective cloud connectivity and data visualization, it did not incorporate advanced analytics or predictive modeling for proactive system management [6].

1.2 Research Gap

Despite rapid progress in IoT-enabled NFT hydroponics, the literature still lacks adaptive control, water reuse, robust calibration, predictive analytics, end-to-end integration, and Capsicum-specific optimization, which directly motivates the proposed inorganic NFT system for chili cultivation [3], [12], [14], [15], [20].

Specialized Design for Capsicum Chili Cultivation

Capsicum chili demands meticulous management of nutrient levels, pH, and water quality, given its sensitivity to variations in these elements. Many current hydroponic systems are designed for general use or are optimized for more demanding crops, 10 failing to provide the specific adjustments and monitoring features required for chili plants [5], [10]. This system is designed to concentrate on capsicum chili, employing real-time monitoring and fuzzy logic-driven automation to regulate nutrient delivery, pH balancing, and water flow according to the plant's changing requirements. The tailored nutrient solutions formulated for chili plants play a vital role in maintaining optimal growth conditions, effectively filling a significant gap in the design of hydroponic systems specific to crops.

Lack of Adaptive Control and Integration of Fuzzy Logic

Many studies into hydroponic systems utilizing IoT emphasize the observation of environmental factors such as pH, EC, and temperature. However, they frequently fall short in providing dynamic and adaptive management of nutrients and pH informed by real-time data. These systems generally depend on threshold-based control or manual adjustments, which do not adequately consider the varying requirements of plants throughout their growth stages. The ongoing investigation examined the application of fuzzy logic within hydroponic systems, concentrating mainly on the regulation of water flow and temperature, while largely overlooking aspects such as nutrient management and real-time pH adjustments [2], [11]. In contrast, the solution we propose integrates real-time fuzzy logic control for the adaptive management of nutrients and pH levels. Through the dynamic adjustment of nutrient levels based on sensor inputs and growth stages, this system aims to improve nutrient efficiency and plant health, tackling a notable limitation in existing systems that fail to respond to real-time data [2].

Insufficient Growth Tracking and Predictive Analytics

A notable constraint in current systems is the absence of growth tracking and predictive analytics. Several studies have incorporated sensor-based monitoring systems for environmental control. However, there is limited scope of growth tracking to dynamically adjust nutrient dosing [1], [4], [5]. Numerous systems depend on fundamental height measurements to track plant growth, yet they often fail to fully incorporate this data into the control system for optimizing nutrient delivery. The lack of growth forecasting utilizing real-time data restricts the system's capacity to enhance plant yield progressively [9].

Our project addresses this gap by incorporating ultrasonic/ToF sensors for real-time monitoring of plant height and linking it with cloud analytics to forecast growth. This system will offer valuable insights into the plant's development, allowing for dynamic adjustments to the nutrient solution, thereby ensuring that optimal growth conditions are upheld throughout the plant's life cycle [2], [3].

Limited Focus on Water and Nutrient Recycling

Water management is a critical issue in hydroponic farming, particularly in water-scarce areas like Sri Lanka. However, numerous studies overlook the incorporation of sustainable practices, such as water recycling. The effectiveness of hydroponic systems in utilizing water has been established [1], [7], yet the aspects of water recycling and nutrient solution filtration are rarely addressed in relation to inorganic NFT systems. Many current systems fail to integrate water recycling or rely on ineffective methods that continue to lead to excessive water usage [3].

The system effectively combines water and nutrient recycling via in-line filtration, leading to a reduction in water waste while simultaneously preserving the quality of the nutrient solution over extended durations. This guarantees that the system functions in a more sustainable manner, minimizing the negative environmental impact of hydroponic agriculture.

Insufficient Long-Term Sensor Calibration and reliability

The accuracy and reliability of sensors are essential for the sustained effectiveness of hydroponic systems. However, numerous IoT-based solutions overlook the critical aspect of sensor calibration. Investigations highlight the significance of sensor calibration. However, many systems continue to depend on manual calibration, which may lead to errors as time progresses [2], [4]. Moreover, environmental factors can cause sensor drift, resulting in inaccurate measurements and diminished system performance. To tackle this issue, our system will utilize semi-automated sensor calibration to guarantee that sensors uphold long-term accuracy and deliver dependable data. This groundbreaking method greatly enhances the dependability and uniformity of the system, minimizing the necessity for human involvement and extending the lifespan of the system.

Gaps in Remote Monitoring and Control Systems

In the end, although cloud-based dashboards are gaining popularity in hydroponic systems, there remains a scarcity of systems that provide extensive remote controls. Current platforms mainly emphasize data visualization, which restricts their capacity to implement real-time modifications to nutrient levels, pH, and water flow [2], [3], [4]. The system addresses this limitation by offering cloud-based real-time data visualization and remote control through an intuitive dashboard. This enables operators to modify system parameters from a distance, ensuring that optimal conditions are maintained even in the absence of physical human interaction.

Table 1.1: Gaps and Proposed System

Feature	Current Limitations	Proposed Innovation
Capsicum Chili Focus	Few studies target capsicum chili, using generic setups for crops like lettuce [1], [3], [10], [20].	Tailored NFT system for capsicum chili, optimizing pH and EC for 20-30% better yields [17].
Real-Time Monitoring	Basic monitoring (pH, EC) with limited IoT integration, risking delays [2], [4], [15].	Full IoT monitoring with ESP32 sensors (pH, EC, water level) and cloud dashboard for instant control [1], [3].
Intelligent Control	Static or basic controls (threshold-based) cause nutrient imbalances [1], [14].	Real-time fuzzy logic on ESP32 for dynamic nutrient and pH adjustments, improving growth [17].
Water Recycling	No water reuse, wasting 20-30% water and raising costs [3], [10].	In-line filtration and recycling system, cutting water use by 15-20% for sustainability [3].
Sensor Calibration	Manual calibration leads to errors over time [7], [12], [14].	Semi-automated calibration via dashboard, ensuring reliable long-term accuracy [12].
Growth Monitoring	Basic visual checks, missing real-time data [1], [4], [9].	Ultrasonic/ToF sensors with cloud analytics for real-time plant height tracking.

1.3 Research Problem

Capsicum chili cultivation in Sri Lanka faces mounting challenges from water scarcity, unpredictable climate patterns, and declining yields that threaten food security and agricultural livelihoods. Traditional soil-based farming consumes excessive water resources and struggles with nutrient management, making hydroponic NFT systems an attractive alternative for water-efficient cultivation, yet current inorganic NFT systems for Capsicum chili suffer from critical limitations that prevent optimal yield achievement and resource utilization [3], [10]. Existing automated hydroponic systems rely predominantly on static, threshold-based control mechanisms that fail to adapt to dynamic growing conditions and plant physiological needs, leading to suboptimal growing conditions particularly problematic for Capsicum species, which exhibit salt sensitivity and require tighter pH control (5.8-6.2) and EC management (1.8-2.2 mS cm^{-1}) compared to more tolerant crops [7], [10], [20]. IoT-enabled monitoring systems, while providing real-time data collection, prioritize visualization over adaptive decision-making, with current cloud platforms lacking predictive analytics capabilities and failing to translate sensor data into proactive control actions, missing opportunities for yield optimization, while sensor calibration drift significantly undermines long-term system reliability, with pH and EC probe accuracy deteriorating over weeks of continuous operation [1], [7], [9], [15]. Water sustainability remains a major gap, as most NFT implementations lack integrated water reuse and filtration systems, resulting in 20-30% water wastage, a limitation that contradicts the fundamental advantage of hydroponic cultivation and increases operational costs in water-scarce regions like Sri Lanka [10], [20]. These converging challenges result in yield losses of up to 20-30% compared to optimal growing conditions, undermining the economic viability of hydroponic farming for smallholder farmers and commercial operations, with the lack of integrated solutions that address adaptive control, water reuse, sensor reliability, and crop-specific optimization simultaneously limiting the scalability of sustainable Capsicum chili production.

1.4 Research Questions

RQ1: How can an integrated NFT system with semi-automated sensor calibration maintain long-term measurement accuracy for optimal Capsicum chili environmental monitoring?

RQ2: To what extent can real-time fuzzy logic control improve nutrient efficiency and pH stability compared to conventional threshold-based automation in Capsicum chili cultivation?

RQ3: How effectively can integrated water reuse with predictive growth monitoring reduce resource consumption while maintaining optimal growing conditions for Capsicum chili?

RQ4: What impact do cloud-based predictive analytics have on yield optimization and decision-making efficiency in automated hydroponic Capsicum chili production?

2. OBJECTIVES

2.1 Main Objectives

To design, develop and implement an energy-efficient automated inorganic NFT hydroponic system specifically optimized for Capsicum chili cultivation, incorporating real-time fuzzy logic control, semi-automated sensor calibration, water recycling, and cloud-based predictive analytics to enhance yield sustainability and resource efficiency under Sri Lankan agricultural conditions.

2.2 Specific Objectives

1. To construct an integrated inorganic NFT hydroponic infrastructure with semi-automated sensor calibration capabilities

Design and build a complete NFT hydroponic system comprising nutrient tank, grow channels, water circulation system, and ESP32-based sensor network (pH, EC, temperature, humidity, ultrasonic/ToF). Implement semi-automated sensor calibration protocols to maintain long-term measurement accuracy and ensure reliable environmental parameter monitoring throughout the cultivation cycle. The system will be tailored to address Capsicum chili's salt sensitivity and specific nutrient requirements within the inorganic growing medium.

2. To implement real-time fuzzy logic-based automation for adaptive nutrient and pH control

Develop and deploy a Mamdani fuzzy logic control system on ESP32 microcontroller to manage dynamic nutrient dosing, pH adjustment, and water flow based on real-time sensor inputs and growth-stage requirements. The fuzzy control algorithm will eliminate static threshold-based limitations by continuously adapting to changing environmental conditions and plant physiological needs, providing responsive nutrient management specifically calibrated for Capsicum chili cultivation phases.

3. To integrate sustainable water reuse and filtration mechanisms with predictive growth monitoring

Implement an inline filtration and water recirculation system to minimize water waste while maintaining nutrient solution quality over extended periods. Deploy ultrasonic/ToF sensors for non-destructive, real-time monitoring of plant height and growth progression, enabling predictive adjustments to optimize growing conditions and resource utilization throughout the Capsicum chili life cycle.

4. To develop a cloud-based dashboard for real-time monitoring, predictive analytics, and remote system management

Create a Firebase-hosted platform for comprehensive real-time data visualization, growth trend analysis, and predictive yield forecasting using integrated sensor data streams. The dashboard will provide remote monitoring capabilities, automated alert systems for parameter deviations, and predictive analytics to support proactive decision-making and continuous optimization of the hydroponic growing environment.

3. METHODOLOGY

3.1 System Diagram

The system diagram below (Figure 3.1) illustrates the components and how they work together to achieve the main objective of this project: optimizing capsicum chili growth through an automated inorganic NFT hydroponic system. The key components include the NFT hydroponic setup, ESP32 microcontroller, sensors, fuzzy logic automation, and cloud dashboard. Each component has been chosen to work cohesively, enabling real-time nutrient and pH control, growth tracking, and sustainable water usage.

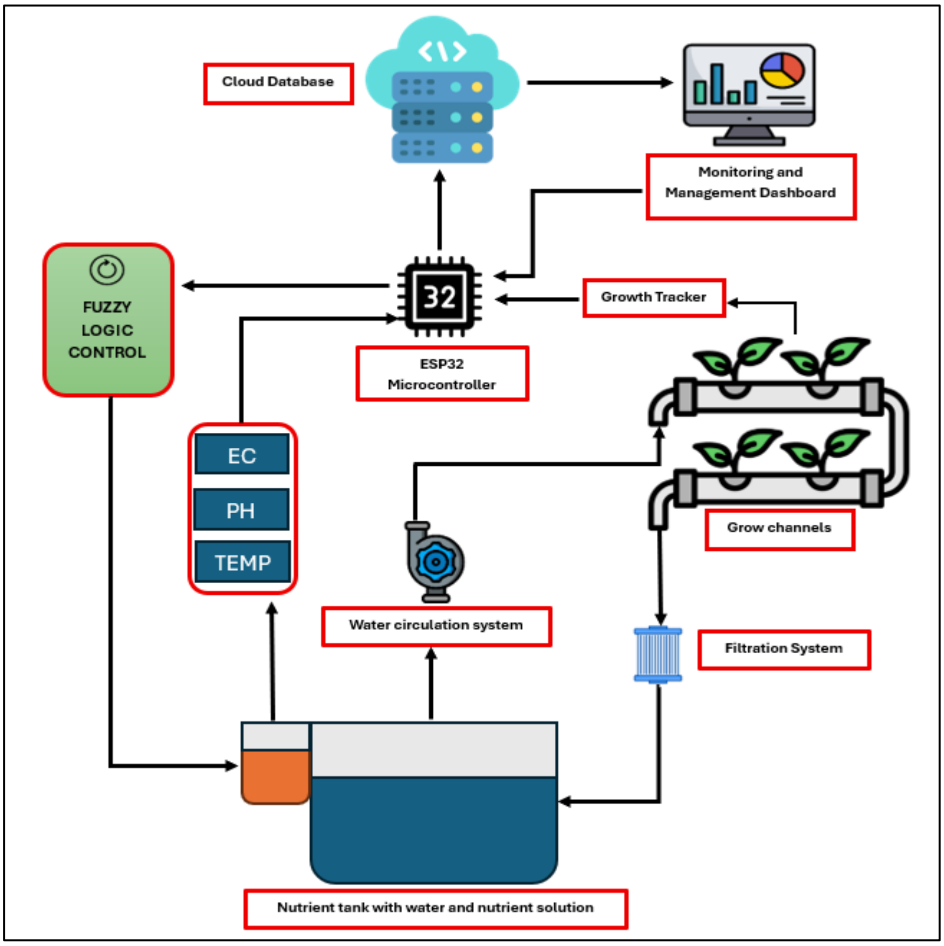

Figure 3.1: System Diagram

The system proposed is an automated inorganic NFT (Nutrient Film Technique) hydroponic solution for Sri Lankan capsicum chili farming under Sri Lankan climatic conditions. The overall function system architecture (Figure 3.1) combines IoT-based sensing, real-time control algorithms, and cloud-based analytics for intelligent nutrient and pH control, sustainable water reuse, and remote monitoring.

The hydroponic system will be built with a nutrient reservoir linked to NFT grow channels via a water circulation system powered by submersible pumps. The setup guarantees a continuous flow of a thin film of nutrient solution over plant roots, delivering necessary nutrients and oxygen while reducing wastage of water. For increased sustainability, the system has a closed-loop water recycling system coupled with an in-line filtration module for the elimination of impurities and preservation of nutrient solution integrity upon recirculation.

At the heart of the control system is the ESP32 microcontroller, selected due to its dual-core processing capability, integral Wi-Fi, and low power usage, which is well-suited for IoT-based agricultural systems. The ESP32 will be coded in C++ within the Arduino IDE, utilizing lightweight IoT communication as well as HTTP protocols for cloud synchronization of data. Sensor data will be processed in real time locally to provide instantaneous responses with continued remote access through the cloud.

The sensor layer is comprised of pH and Electrical Conductivity (EC) sensors for acidity level and nutrient concentration monitoring, temperature and humidity sensors for environmental monitoring, and ultrasonic or Time-of-Flight (ToF) sensors for growth monitoring in a non-destructive manner. The integration of sensors will be based on a calibrated installation procedure, with semi-automated calibration routines being integrated into the ESP32 firmware to provide long-term accuracy of measurements with little manual intervention. Smart control of nutrient dosing, pH balancing, and water flow will be implemented using fuzzy logic algorithms integrated into the microcontroller. In contrast to static threshold-based systems, fuzzy logic will facilitate adaptive decision-making from real-time information, so that changes in environmental conditions or plant nutrient uptake will be managed dynamically. Peristaltic pumps and solenoid valves will be operated based on fuzzy inference results, providing accurate adjustments to the nutrient solution without wastage of resources.

For remote monitoring and control, the system will be coupled to a cloud-based dashboard built on Google Firebase that offers a secure environment for real-time data visualization and system analytics. The dashboard interface will be created using web technologies (HTML, CSS, JavaScript) for viewing important metrics like pH, EC, temperature, humidity, and plant growth trends. Other features will be historical data logging, automatic alerts for threshold violations, and manual override controls for system safety and user flexibility. A mobile-friendly interface will provide accessibility for farmers in rural and semi-urban areas.

Lastly, for system reliability and performance, there will be periodic testing and optimization cycles throughout implementation. This encompasses calibration verification, nutrient dosing verification, and fuzzy logic performance testing under dynamic environmental conditions. This methodical process ensures a solid, scalable, and power-efficient hydroponic solution that maximizes capsicum chili yield while mitigating the limiting factors of water and resource availability in Sri Lanka.

3.2 Time Frame

The project will follow a structured timeline to ensure all phases are completed efficiently and on schedule. The key tasks and their durations are outlined below, providing a clear roadmap for the project's progression from research to final documentation.

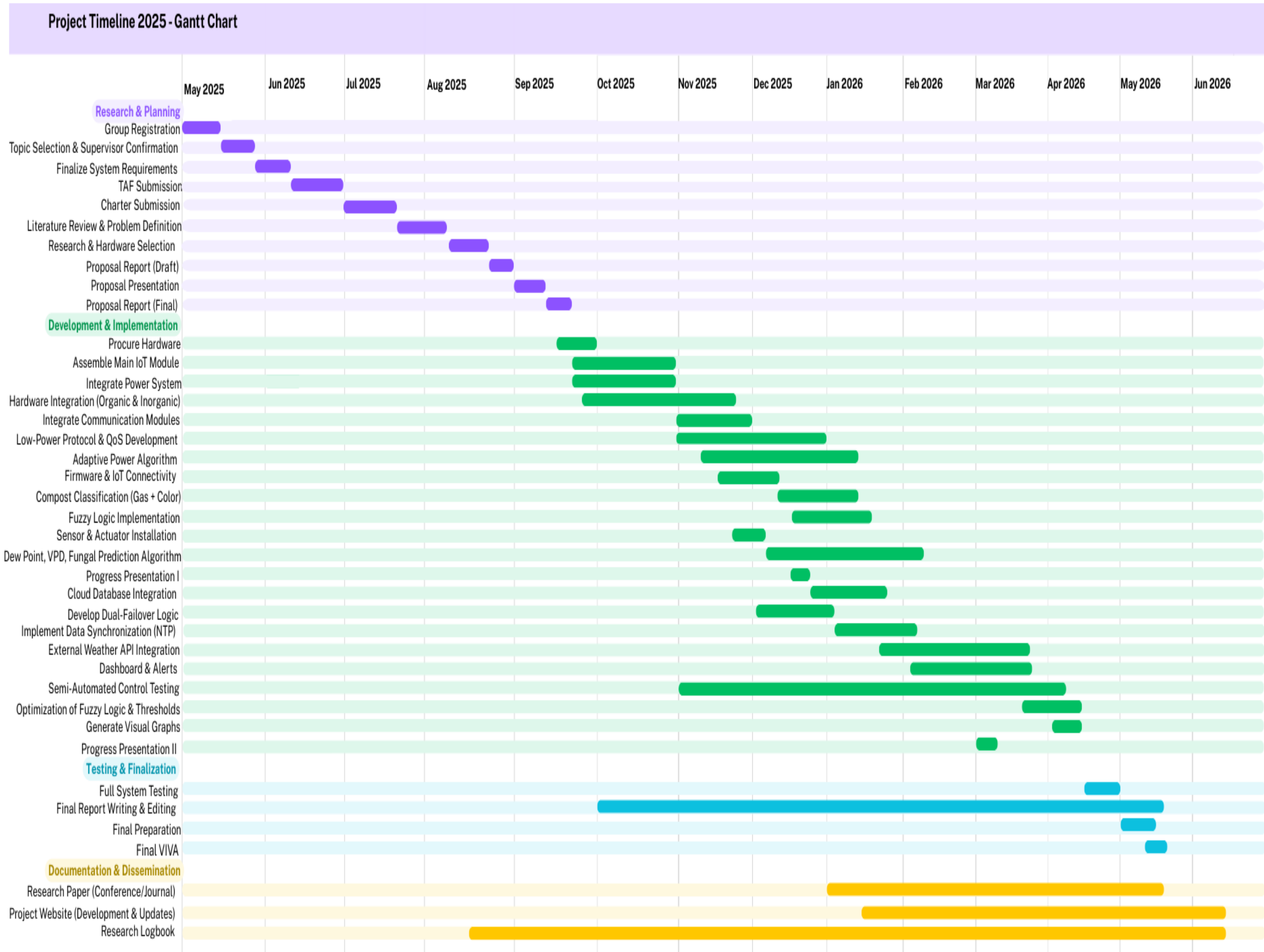

Figure 13.2: Grant Chart

3.3 Work Breakdown Structure: Inorganic NFT Hydroponic System

The subsystem will be delivered over five phases from July 24, 2025, to May 20, 2026, ensuring milestones are met for efficient, high-reliability completion.

Phase 1: System Design & Planning (July – September 2025)

- Analyze project requirements for the inorganic NFT subsystem.
- Specify ESP32, required sensors (pH, EC, temperature, humidity, ultrasonic/ToF), and actuators.
- Complete literature review and finalize electronic, mechanical block diagrams for proposal.

Phase 2: Hardware Setup & Sensor Integration (September – November 2025)

- Design NFT hydroponic layout: nutrient tank, channels, pump, filtration circuit.
- Procure all hardware: PVC channels, tank, sensors, ESP32, pumps, filtration components.
- Assemble the NFT system and install all sensors with wiring to ESP32.
- Calibrate all sensors (pH, EC, temp, humidity, ultrasonic/ToF).
- Develop and test initial ESP32 firmware (Arduino IDE, C++) for reliable sensor data acquisition.
- Establish initial water circulation and filtration loop.

Phase 3: Firmware, Cloud & Fuzzy Control (November – December 2025)

- Develop cloud connectivity for data upload (Firebase integration).
- Implement fuzzy logic for adaptive nutrient dosing, pH balancing, water flow.
- Test automation features using real sensor data and actuators.
- Enable remote device communication and data monitoring for troubleshooting.

Phase 4: Advanced Features & Optimization (January – March 2026)

- Refine fuzzy logic and add predictive analytics to cloud dashboard.
- Integrate semi-automated sensor calibration system for reliability.
- Optimize water reuse and filtration for efficiency and sustainability goals.
- Implement dashboard features allowing manual override and real-time system alerts.

Phase 5: Validation, Evaluation & Reporting (March – May 2026)

- Conduct final system stress tests using capsicum plants.
- Benchmark performance vs. traditional operation (yield, energy, water use).
- Document calibration, maintenance, and troubleshooting procedures.
- Prepare subsystem documentation for main report, evaluation, and viva support.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

1. Real-Time Environmental and Nutrient Monitoring

- The system must continuously monitor key parameters such as pH, electrical conductivity (EC), temperature, humidity, and water level through integrated sensors.
- Data from the sensors should be transmitted in real-time to the ESP32 microcontroller for processing.

2. Automated Nutrient and pH Control

- The system should automatically control the nutrient dosing, pH balancing, and water flow based on real-time sensor data using fuzzy logic automation.
- The fuzzy logic control system must adjust nutrient solutions to ensure optimal growth conditions for capsicum chili, minimizing resource waste.

3. Water Reuse and Filtration

- The system must implement a water recycling mechanism that continually recirculates the nutrient solution, ensuring efficient use of water.
- An in-line filtration system should be integrated to maintain water purity by removing contaminants before the nutrient solution is returned to the nutrient tank.

4. Cloud-Based Data Visualization

- The system should be able to send data to a cloud platform for real-time visualization of parameters such as nutrient levels, pH, EC, water temperature, and growth trends.
- Users should be able to access the dashboard remotely via web and mobile applications, enabling real-time monitoring and decision-making.

5. Sensor Calibration and Growth Tracking

- The system must support semi-automated sensor calibration to ensure accurate long-term monitoring.
- Ultrasonic/ToF sensors should be used to track plant height and yield progression in a non-destructive manner.

6. Alerts and Notifications

- The system should generate alerts and notifications when parameters exceed predefined thresholds, prompting users to take corrective actions.

4.2 Non-functional Requirements

1. Reliability

- The system must be highly reliable, providing continuous operation without frequent failures or downtime.
- Redundant systems and automated calibration should ensure that the hydroponic system functions efficiently over extended periods.

2. Scalability

- The system should be scalable, allowing for expansion to larger farms or multiple hydroponic units.
- It should support the addition of more sensors and controllers without significant changes to the core system architecture.

3. Usability

- The user interface (UI) for the cloud-based dashboard should be intuitive, user-friendly, and easy to navigate for both technical and non-technical users.
- The system must include clear visualizations of sensor data, alerts, and plant growth trends to facilitate decision-making.

6. Performance

- The system should provide fast, real-time processing of sensor data and responsive adjustments to nutrient and pH levels.

7. Sustainability

- The system must be designed with sustainability in mind, particularly regarding water usage and energy consumption. It should minimize resource waste through efficient water recycling and low-energy operations, especially in off-grid environments.

4.3 User Requirements

User requirements define the needs of the stakeholders who will interact with the system, such as farm operators, technicians, and remote managers.

1. Remote Access and Control

- Users should be able to monitor and control the system remotely via a cloud-based dashboard accessible from devices.
- The dashboard should allow users to view real-time sensor data, adjust settings, and receive alerts and notifications.

2. Automated System with Minimal Intervention

- The system must be capable of automating the nutrient delivery, pH balancing, and water circulation processes, reducing the need for manual interventions and allowing for hands-off operation.
- The ability to override the automated system remotely should be available for users in case of emergencies or adjustments.

3. Simple Setup and Configuration

- The system should be easy to set up and configure, with step-by-step installation guides and automated calibration to minimize installation time.
- Initial configuration of sensors, water circulation, and nutrient dosing parameters should be straightforward and guided by clear instructions.

4. Data Reporting and Analytics

- Users should be able to access historical data, analyse trends related to plant growth, nutrient levels, water usage, and system performance.
- The system should support customizable analytics to help users make informed decisions about plant care and system optimization.

4.4 System Requirements

4.4.1 Hardware Requirements

- **ESP32 Microcontroller** – Serves as the central processing and decision-making unit. It manages sensor data acquisition, executes fuzzy logic algorithms, and controls actuators. Chosen for its built-in Wi-Fi, low power consumption, and compatibility with IoT protocols.
- **pH Sensor** – Monitors the acidity and alkalinity of the nutrient solution to ensure optimal conditions for nutrient absorption.
- **Electrical Conductivity (EC) Sensor** – Tracks the nutrient concentration within the solution to maintain precise feeding levels for capsicum growth.
- **Temperature & Humidity Sensor (DHT22/SHT31)** – Measures ambient conditions that influence evapotranspiration and plant metabolism.
- **Ultrasonic Water Level Sensor** – Detects the level of nutrient solution in the reservoir, preventing pump dry-run and optimizing water management.
- **Ultrasonic/Time-of-Flight (ToF) Sensor** – Enables non-invasive, real-time measurement of plant height for growth tracking and yield estimation.
- **Peristaltic Pumps and Solenoid Valves** – Used for automated dosing of nutrients and pH control solutions with precision, based on fuzzy logic decisions.
- **Relays & Power Modules** – To control high-voltage components such as pumps and solenoid valves safely through the microcontroller.

- **Nutrient Tanks and NFT Grow Channels** – A structured hydroponic layout for circulating nutrient solution efficiently across capsicum chili roots.
- **Filtration Unit for Water Reuse** – Integrated in-line filter for recycling the nutrient solution, maintaining purity and reducing water wastage.

4.4.2 Software Requirements

- **Embedded Programming Environment** – Arduino IDE for writing and uploading firmware to the ESP32 microcontroller.
- **Programming Languages** – C++ (for embedded control logic) and Arduino libraries for IoT sensor integration and fuzzy logic implementation.
- **Automation Algorithms** – Fuzzy Logic Engine embedded on the ESP32 for intelligent nutrient dosing, pH balancing, and adaptive water flow control based on real-time sensor feedback.
- **Communication Protocols** – MQTT for lightweight IoT communication between the ESP32 and cloud, and HTTP/REST APIs for updating cloud-based databases.
- **Cloud Platform** – Google Firebase for real-time data storage, authentication, and synchronization between the device and user interface.
- **Dashboard Application** – A cloud-connected web dashboard for live visualization of pH, EC, water level, temperature, and plant growth metrics, including manual override controls.
- **Data Analytics** – Includes trend analysis for nutrient efficiency, plant growth monitoring, and predictive adjustments for crop cycles.
- **Mobile Access** – Responsive web interface optimized for smartphones, enabling remote monitoring and control for farmers.
- **Notification System** – Cloud-based alerting mechanism to notify users of parameter deviations, pump failures, or sensor calibration requirements.

4.5 Expected Test Cases

The following test cases are designed to validate the functionality, performance, and reliability of the proposed automated inorganic NFT hydroponic system:

1. Sensor Accuracy Test

- **Objective** - Verify that all sensors (pH, EC, temperature, humidity, water level, ultrasonic/ToF) provide accurate readings.
- **Method** - Compare sensor readings against standard calibrated devices.
- **Expected Result** - Readings should be within ±5% of the reference values.

2. Fuzzy Logic Control Response Test

- **Objective** - Evaluate the accuracy and responsiveness of the fuzzy logic algorithm for nutrient dosing and pH adjustment.
- **Method** - Introduce controlled variations in pH and EC levels and observe system adjustments.
- **Expected Result** - The system should stabilize pH and EC within predefined optimal ranges.

3. Real-time Data Transmission Test

- **Objective -** Validate the IoT-based data communication between ESP32 and the cloud platform.
- **Method** - Simulate real-time sensor data transmission and monitor updates on the dashboard.
- **Expected Result** - Data should be updated on the dashboard without packet loss.

4. Cloud Dashboard Functionality Test

- **Objective** - Ensure accurate visualization and control through the cloud dashboard.
- **Method** - Perform manual override commands via the dashboard.
- **Expected Result** - Commands should execute successfully with real-time feedback.

5. Water Recycling and Filtration Efficiency Test

- **Objective** - Assess the effectiveness of the water reuse mechanism and in-line filtration system.
- **Method** - Measure water clarity and nutrient retention before and after filtration during recirculation cycles.
- **Expected Result** - Filtration should maintain nutrient solution quality and clarity for reuse.

6. Sensor Calibration Functionality Test

- **Objective** - Test semi-automated calibration routines for long-term accuracy.
- **Method** - Trigger calibration remotely and verify updated baseline readings.
- **Expected Result** - Sensors should return to accurate readings after calibration.

7. Plant Growth Tracking Accuracy Test

- **Objective -** Validate ultrasonic/ToF sensor measurements for plant height tracking.
- **Method** - Compare sensor-measured height with manual measurement at different growth stages.
- **Expected Result** - Accuracy within of actual plant height.

8. Alert and Notification Test

- **Objective** - Check whether the system generates timely alerts when parameters exceed thresholds.
- **Method** - Intentionally alter pH, EC, or water levels beyond the safe range and monitor alerts.
- **Expected Result** - Alerts should trigger parameter deviation.

9. System Reliability and Uptime Test

- **Objective** - Assess continuous system operation without failure.
- **Method** - Run the system continuously under controlled conditions.
- **Expected Result** - No system crashes, real-time monitoring and control remain functional throughout the test period.

10. Energy Efficiency Test

- **Objective** - Evaluate energy consumption of pumps, sensors, and controllers.
- **Method** - Measure total energy usage for 24-hour operation.
- Expected Result - Energy consumption should remain within expected limits for small-scale NFT hydroponic systems.

5. BUDGET AND BUDGET JUSTIFICATION

Table 25.1: Budget and Justification

Component	Amount (LKR)	Justification
ESP32 Microcontroller	4,000	Core controller for sensor data processing and system automation.
pH Sensors	2,000	Monitors acidity of the nutrient solution, ensuring optimal nutrient uptake for capsicum chili.
EC (Electrical Conductivity) Sensors	2,000	Measures nutrient concentration, ensuring proper fertilization for plant growth.
Temperature Sensors (DHT22)	2,000	Monitors water and air temperature to maintain optimal growth conditions.
Humidity Sensors (DHT22)	2,000	Ensures moisture control in the growing area, preventing water stress or Mold.
Ultrasonic/ToF Sensors	5,000	Tracks plant growth without damaging plants.
Water Pumps	6,000	Circulates nutrient solution for optimal nutrient and oxygen delivery to plant roots.
Filtration System	8,000	Filters nutrient solution, ensuring cleanliness and system efficiency.
Water Tanks	8,000	Holds nutrient solution for circulation, ensuring continuous water supply to plants.
Nutrient Solution	5,000	Purchases essential nutrients (e.g., nitrogen, phosphorous) required for plant growth.
Miscellaneous Components	5,000	Covers small parts such as wires, connectors, and mounting materials necessary for system assembly.

- **Total estimate budget: LKR 49,000.00**

6. CONCLUSION

This study proposal outlines an innovative strategy for addressing the urgent issues of food insecurity, climatic variability, and resource constraint in Sri Lanka by creating an energy-efficient dual hydroponic system for both organic and inorganic capsicum chilli cultivation. The proposed system addresses critical shortcomings in current inorganic Nutrient Film Technique (NFT) configurations, including dependence on static threshold-based automation that does not adjust to the changing requirements of plants, insufficient water recycling resulting in considerable waste, and inconsistent manual sensor calibration susceptible to drift, thereby promoting sustainable precision agriculture through innovative integrations. It utilizes real-time fuzzy logic control for adaptive nutrient dosing and pH regulation, cloud-based analytics with ultrasonic/ToF sensors for non-invasive growth monitoring and predictive analysis, in-line filtration and sterilization for effective water/nutrient recycling, and semi-automated calibration processes to guarantee sustained data precision and reduced intervention.

This scalable approach aims for a 20-30% increase in yield and a 15-20% decrease in water usage, optimizing capsicum chili growth in off-grid, resource-limited settings while reducing environmental impact and operational expenses. The research offers actionable insights into the trade-offs among efficiency, sustainability, and crop quality through comparative evaluations of organic and inorganic modules. This initiative enhances precision agriculture by providing a replicable model that empowers small-scale farmers, strengthens economic resilience, and supports Sri Lanka's sustainable development goals, facilitating the wider adoption of intelligent hydroponic technologies in developing areas.

REFERENCE

[1] S. Shaha and R. K. Budhathoki, "IoT - Driven Automation and Monitoring in Hydroponic Lettuce Farming," *2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India, 2024, pp. 1544-1549, doi: 10.1109/ICACCS60874.2024.10717081.

[2] Y. V. Bhargava, P. K. Chittoor, C. Bharatiraja, R. Verma and K. Sathiyasekar, "Sensor Fusion Based Intelligent Hydroponic Farming and Nursing System," in *IEEE Sensors Journal*, vol. 22, no. 14, pp. 14584-14591, 15 July15, 2022, doi: 10.1109/JSEN.2022.3177777.

[3] A. Mendon, B. M. Votavat and S. Singh, "Design and Construction of Automated Hydroponics System with Remote Monitoring and Control," *2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, Kharagpur, India, 2022, pp. 1-5, doi: 10.1109/ICCCNT54827.2022.9984617.

[4] R. Basuel, A. Fajardo and N. D. Perez, "Design and Development of an IoT-Based Plant Growth Monitoring System Using Esp-32 Cam," *2024 IEEE International Conference on Control & Automation, Electronics, Robotics, Internet of Things, and Artificial Intelligence (CERIA)*, Bandung, Indonesia, 2024, pp. 1-4, doi: 10.1109/CERIA64726.2024.10914805.

[5] A. M. Zaid *et al*., "IoT-based Smart Hydroponic System Using Nutrient Film Technique (NFT) for Lettuce Plant," *2023 IEEE 12th Global Conference on Consumer Electronics (GCCE)*, Nara, Japan, 2023, pp. 976-980, doi: 10.1109/GCCE59613.2023.10315640.

[6] Rajaseger G, Chan KL, Yee Tan K, Ramasamy S, Khin MC, Amaladoss A, Kadamb Haribhai P. Hydroponics: current trends in sustainable crop production. Bioinformation. 2023 Sep 30;19(9):925-938. doi: 10.6026/97320630019925. PMID: 37928497; PMCID: PMC10625363.

[7] C. Suresh and S. S, "Enhancing Hydroponic Farming Using IoT Integration and Machine Learning Techniques," *2025 International Conference on Multi-Agent Systems for Collaborative Intelligence (ICMSCI)*, Erode, India, 2025, pp. 439-444, doi: 10.1109/ICMSCI62561.2025.10894677.

[8] M. Wildan and N. Anisa, "Internet of Things System Development and Experimentation Hydroponic Farming," *2024 International Conference on Information Management and Technology (ICIMTech)*, Bali, Indonesia, 2024, pp. 54-59, doi: 10.1109/ICIMTech63123.2024.10780827.

[9] R. Lal, "Autonomous Hydroponic Farming using Internet of Things," 2024 1st International Conference on Cognitive, Green and Ubiquitous Computing

(IC-CGU), Bhubaneswar, India, 2024, pp. 1-6, doi: 10.1109/IC-CGU58078.2024.10530766.

[10] P. Kalyan, R. Surendran and R. Selvanarayanan, "Hydroponic NFT for Brinjal cultivation using Decision Tree (DT) and K-Nearest Neighbour (KNN)," *2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)*, Erode, India, 2024, pp. 233-238, doi: 10.1109/ICSSAS64001.2024.10760940.

[11] M. N. T. Perera *et al.*, "Intelligent Algorithm for Optimizing Hydroponic Solution in IoT-Integrated Agriculture Systems," *2024 Moratuwa Engineering Research Conference (MERCon)*, Moratuwa, Sri Lanka, 2024, pp. 133-138, doi: 10.1109/MERCon63886.2024.10689134.

[12] M. S. A. B. M. Ali, M. K. B. Nordin, M. H. B. M. Zaki and M. F. B. Saaid, "Optimizing Plant Growth in Indoor NFT Hydroponic Systems: Design, Environmental Monitoring, and Analysis," *2024 IEEE International Conference on Applied Electronics and Engineering (ICAEE)*, Shah Alam, Malaysia, 2024, pp. 1-6, doi: 10.1109/ICAEE62924.2024.10667552.

[13] N. C. Sayekti, M. A. Fhaizal, W. Wijaya, N. Hendajany, I. Aprianti and Kusmadi, "Smart Farming Design with NodeMCU ESP32 for NFT HYDROPONIC," *2024 18th International Conference on Telecommunication Systems, Services, and Applications (TSSA)*, Bali, Indonesia, 2024, pp. 1-4, doi: 10.1109/TSSA63730.2024.10864065.

[14] G. Gokulraj, R. N. D. Kumar, M. S. Kumar, K. M. Kirthika, B. Vadivel and V. Padmacharan, "Smart IoT based Nutrition Alert System for Hydroponic Plant Growth," *2025 5th International Conference on Trends in Material Science and Inventive Materials (ICTMIM)*, Kanyakumari, India, 2025, pp. 1154-1160, doi: 10.1109/ICTMIM65579.2025.10988447.

[15] E. Y. Fitria, M. A. Murti and C. Setianingsih, "Design of Integrated Control System Based On IoT With Context Aware Method In Hydroponic Plants," *2021 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology (IAICT)*, Bandung, Indonesia, 2021, pp. 20-26, doi: 10.1109/IAICT52856.2021.9532550.

[16] Y. Zheng, X. Hui, D. Cai, M. R. Shoukat, Y. Wang, Z. Wang, F. Ma, and H. Yan, “Calibrating ultrasonic sensor measurements of crop canopy heights: a case study of maize and wheat,” *Frontiers in Plant Science*, vol. 15, 2024. doi:10.3389/fpls.2024.1354359.

[17] J. C. V. Puno, J. J. I. Haban, J. D. Alejandrino, A. A. Bandala and E. P. Dadios, "Design of A Nutrient Film Technique Hydroponics System with Fuzzy Logic Control," *2020 IEEE REGION 10 CONFERENCE (TENCON)*, Osaka, Japan, 2020, pp. 403-408, doi: 10.1109/TENCON50793.2020.9293749.

[18] Fitriani, Z. Zainuddin and Syafaruddin, "Nutrition Control System In Nutrient Film Technique (NFT) Hydroponics With Convolutional Neural Network (CNN) Method," *2022 2nd International Seminar on Machine Learning, Optimization, and Data Science (ISMODE)*, Jakarta, Indonesia, 2022, pp. 41-46, doi: 10.1109/ISMODE56940.2022.10180412.

[19] W. Kazmi, S. Foix, and G. Alenyà, "Plant leaf imaging using time of flight camera under sunlight, shadow and room conditions," in *Proc. IEEE Int. Symp. Robotic and Sensors Environments (ROSE)*, 2012, pp. 192–197. doi: 10.1109/ROSE.2012.6402615.

[20] R. Erel, T. T. Le, A. Eshel, S. Cohen, R. Offenbach, T. Strijker, and I. Shtein, "Root development of bell pepper (Capsicum annuum L.) as affected by water salinity and sink strength," *Plants*, vol. 9, no. 1, p. 35, Dec. 2019. doi: 10.3390/plants9010035.